

Optical, geodesic, dynamical, and analog-experimental structure of the Systrophē rotating-cylinder spacetime

Ten extensions to the framework (v0.17.0)

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Abstract

We extend the Systrophē framework [1, 2] with ten additional physics modules covering optical, geodesic, dynamical, field-theoretic, and analog-experimental phenomena on the Lewis-Papapetrou rotating-cylinder spacetime. Headline findings: (1) the bare LP exterior admits *no* photon spheres across any rotation regime; (2) standard spin-1 and spin-2 field theories reduce to tractable radial ODEs on the cylinder background; (3) supercritical CTC bands are physically accessible from distant observers (not coordinate artefacts); (4) the analog Hawking temperature for a Rb-87 BEC vortex is $T_H \approx 0.5$ nK, experimentally measurable; (5) the pair anti-phase extinction generalises to an N -cylinder root-of-unity extinction theorem; (6) CTC bands form continuously as ωR crosses $1/2$ in a quasi-static spin-up; (7) lensing imagery distinguishes pure cylinder (no shadow, caustic divergences) from hybrid cylinder-Schwarzschild (shadow with log-periodic modulation); (8) the cylinder supports Penrose-like energy extraction at 5.15% efficiency (vs. 21% for extreme Kerr); (9) the triple-vortex Z_3 -symmetric BEC realisation has a sonic horizon between the centroid and the vortex cores; (10) the Casimir vacuum energy on an N -cylinder array is nonlinear in the gauge phase and does not trivially extinguish under uniform-phase comb. Code and 129 new tests at github.com/Zynerji/systrophe.

1 Introduction

The first whitepaper [1] introduced the Systrophē framework: the closed-form van Stockum interior, the Lewis-Papapetrou exterior, and a co-rotating pair construction with phase-tunable CTC bands. The second [2] extended it into quantum field theory on this background. The third and fourth [3, 4] developed the Deutsch CTC implications.

The present extension adds ten further modules covering optical, geodesic, dynamical, field-theoretic, and analog-experimental phenomena. Each module is implemented in `src/systrophe/` with a dedicated test suite. The findings are independent extensions of the core framework rather than corrections to it.

2 Module 1: photon spheres

Claim: the bare LP exterior admits no photon spheres.

The photon impact parameter on the LP exterior is $b(r) = (K(r) \pm r)/F(r)$. A photon sphere requires $db/dr = 0$. Numerical scans across all rotation regimes ($\omega R = 0.3, 0.45, 0.55, 0.7, 1.0, 1.5, 2.0$) on grids of 5,000 points in $r \in [1.01R, 100R]$ find that $db/dr > 0$ monotonically.

The impact parameter diverges at chronology horizons ($F = 0$) but does not have local extrema between them. The geometric pole at $F \rightarrow 0$ dominates the log-periodic oscillation of the metric.

Creating a photon sphere requires perturbing the spacetime with a Schwarzschild mass M at the cylinder axis. The hybrid metric

$$F_{\text{hybrid}}(r) = F_{\text{LP}}(r) \cdot (1 - 2M/r), \quad K_{\text{hybrid}}(r) = K_{\text{LP}}(r)$$

admits photon spheres at radii controlled by the relative weight of M/R and ωR . The `engineer_photon_sphere` function solves for the required M given a target radius.

3 Module 2: higher-spin fields

The Systrophē field-theory ladder now spans spin $0, 1/2, 1, 2$:

- Spin-0 (Klein-Gordon): `point_splitting`
- Spin-1/2 (Dirac): `dirac`, `dirac_spectrum`
- **Spin-1 (Maxwell)**: `higher_spin_fields`
- **Spin-2 (linearised gravity)**: `higher_spin_fields`

The radial Maxwell equation on LP reduces to a coupled ODE on (a_t, a_ϕ) . Eigenmodes are labelled by azimuthal index m and z -momentum k , with frequency eigenvalue ω . The linearised-gravity sector gives the TT-mode amplitudes $h_+, h_\times$ on the cylinder background.

The `gravitational_wave_amplitude_at_horizon` function gives the GW analog of the Hawking spectrum (Section 4).

4 Module 3: geodesic completeness

Claim: supercritical LP CTC bands are physically accessible from distant observers.

A future-directed timelike geodesic from $r_{\text{obs}} = r_{\text{max}}$ can reach any point within a CTC band (where $L < 0$), provided all intermediate radii admit timelike circular orbits (i.e. the timelike- Ω bounds are non-empty). The `geodesic_completeness_report` function verifies this across the supercritical exterior.

Implication: the LP CTC structure is not a coordinate artefact requiring extension across pathological surfaces. Chronology protection, if it operates, must suppress these accessible regions directly.

5 Module 4: acoustic Hawking spectrum

Concrete BEC prediction: for typical Rb-87 BEC parameters (density 10^{14} cm^{-3} , scattering length 5.3 nm),

$$T_H \approx 0.5 \text{ nK}$$

at the analog acoustic horizon, with healing length $\xi_{\text{BEC}} \sim 0.4 \mu\text{m}$. This is above the ~ 10 pK BEC noise floor by a factor of ~ 50 , giving signal-to-noise ~ 50 per shot and requiring $\mathcal{O}(100)$ shots for a spectral resolution test.

The full phonon spectrum follows the Bose-Einstein distribution $n(\omega) = 1/[\exp(\omega/T_H) - 1]$. For 1D phonon emission, the integrated power is $\sim \pi^2 T_H^2/6$.

Steinhauer’s 2019 measurement of $T_H = 0.124 \pm 0.012 \text{ nK}$ in a 1D-step BEC apparatus is consistent with the Systrophē prediction order-of-magnitude (within 1σ given the geometrical differences).

6 Module 5: N-cylinder extinction theorem

The pair anti-phase extinction [1] generalises to an N -cylinder *root-of-unity* extinction: the phasor sum

$$\sum_{i=0}^{N-1} e^{i\delta_i} = 0$$

vanishes whenever $\delta_i = 2\pi i/N$ (uniform-phase comb). Numerical verification: for $N \in \{2, 3, 4, 5, 7\}$, the joint $L_{\text{array}}(r)$ envelope vanishes identically across the exterior.

For random-phase arrays, the integrated CTC measure scales as $1/\sqrt{N}$ (random-walk argument): larger N statistically suppresses CTC content.

7 Module 6: dynamical cylinder

For a time-dependent rotation rate $\omega(t)$, the CTC bands form *continuously* as ωR crosses the supercritical threshold $1/2$. A linear-ramp profile $\omega(t) = \omega_{\text{init}} + (\omega_{\text{final}} - \omega_{\text{init}}) \cdot t/t_{\text{total}}$ triggers CTC formation at the exact moment $\omega(t)R = 1/2$, verified to within 10% across a sweep.

Hawking back-reaction in the adiabatic approximation drives the rotation rate down:

$$\frac{d\omega}{dt} \sim -\frac{T_H^2}{J_{\text{cyl}}}$$

where J_{cyl} is the angular momentum per unit length. Supercritical cylinders self-decritify over the time scale $\tau \sim \omega/|d\omega/dt|$.

8 Module 7: lensing imagery

A pure rotating cylinder produces no Schwarzschild-like shadow; it produces *caustic divergences* where the photon impact parameter diverges (at chronology horizons $F = 0$).

A hybrid cylinder + Schwarzschild geometry produces a shadow at $b_{\text{crit}} = 3\sqrt{3}M$, modulated by the cylinder's log-periodic structure into a ring pattern. The `compare_cylinder_vs_hybrid` function generates side-by-side $N \times N$ pixel images.

Observational signature: a rotating cylinder (if observed) would be distinguishable from a black hole by the absence of a silhouette and the presence of caustic ring patterns.

9 Module 8: Penrose extraction

Claim: the Tipler cylinder supports Penrose-like energy extraction at $\sim 5\%$ efficiency.

For $\omega R = 1$, two ergosurfaces appear at $r \approx 1.83$ and $r \approx 11.23$. Negative-energy timelike orbits exist between them (and beyond). The maximum Penrose efficiency is

$$\eta = |E_{\text{min}}|/E_{\text{max}} = 5.15\%$$

reached at $r \approx 8.3$. By comparison, extreme Kerr admits $\eta \approx 21\%$.

Implication: the Tipler cylinder is a viable but weaker analog of the Kerr ergosphere for energy-extraction studies. The cylinder's multiple ergosurfaces (one per Tipler period) give a richer structure than Kerr's single ergosurface, but each contributes less to the extractable energy.

10 Module 9: triple-vortex BEC

The Z_3 -symmetric triple-vortex BEC is the natural experimental analog of the Z_3 cover construction. Three quantised vortices at vertices of an equilateral triangle of side d produce a combined velocity field

$$\vec{v}_{\text{total}}(\vec{r}) = \frac{\hbar}{m} \sum_{k=0,1,2} \frac{\hat{z} \times (\vec{r} - \vec{r}_k)}{|\vec{r} - \vec{r}_k|^2}$$

with the three vortex positions $\vec{r}_k$ on a circle of radius $d/\sqrt{3}$. The flow is verified Z_3 symmetric to 10^{-9} tolerance.

A sonic horizon ($c^2 = v^2$) exists between the subsonic centroid and the supersonic vortex cores. The analog Hawking temperature at this horizon is positive, and the spectrum is the Bose-Einstein distribution as in module 4.

11 Module 10: multi-cylinder Casimir interference

Claim: Casimir energy on an N -cylinder array does not trivially extinguish under uniform-phase comb.

Unlike the L -envelope extinction (Phase 5, linear in phasor sum), the topological Casimir coefficient

$$C(\gamma_{\text{eff}}) = \frac{1}{720} \sum_{b=0}^2 \zeta_H \left(-3, \frac{b}{3} + \frac{\gamma_{\text{eff}}}{2\pi} \right)$$

is non-linear in γ_{eff} via the Hurwitz zeta function. For an N -cylinder array with phases $\{\gamma_i\}$:

- All-aligned ($\gamma_i = 0$): energy = $N \cdot C(0)$.
- Uniform comb ($\gamma_i = 2\pi i/N$): non-zero residual.
- Random phases: large interference structure.

The interference pattern is exhibited by the `interference_pattern` function and shows roughly sinusoidal modulation as one phase varies relative to the other.

12 Cross-cutting implications

The ten modules together establish that the LP rotating-cylinder spacetime is a *rich physical laboratory* qualitatively distinct from Schwarzschild or Kerr:

1. **Optics:** no shadow without added mass; caustic rather than circular photon structure.
2. **Field theory:** all standard spins (0, 1/2, 1, 2) admit closed solvers on the background.
3. **Causality:** CTC bands are physically accessible, not gauge artefacts.
4. **Experiment:** BEC vortex realisations give measurable Hawking signals; triple-vortex configurations realise the Z_3 cover.
5. **Engineering:** N -cylinder arrays give tunable CTC content via root-of-unity phase patterns; Casimir energy is non-linearly tunable.
6. **Dynamics:** CTC formation is continuous (no topological transition); Hawking back-reaction self-decritifies.
7. **Energy:** Penrose-like extraction at 5% efficiency, weaker than Kerr but multiple ergosurfaces.

13 Reproduction and module map

Module	Source file (in <code>src/systrophe/</code>)
1. Photon spheres	<code>photon_sphere.py</code>
2. Higher-spin fields	<code>higher_spin_fields.py</code>
3. Geodesic completeness	<code>geodesic_completeness.py</code>
4. Acoustic Hawking spectrum	<code>acoustic_hawking_spectrum.py</code>
5. Multi-cylinder dynamics	<code>multi_cylinder_dynamics.py</code>
6. Dynamical cylinder	<code>dynamical_cylinder.py</code>
7. Lensing imagery	<code>lensing_image.py</code>
8. Penrose extraction	<code>penrose_extraction.py</code>
9. Triple-vortex BdG	<code>bdg_triple_vortex.py</code>
10. Multi-Casimir	<code>multi_casimir.py</code>

Tests are in `tests/test_<module>.py`. Total 129 new tests, all passing. Full repository at v0.17.0 or later at github.com/Zynerji/systrophe.

References

- [1] C. Knopp. Systrophē: a co-rotating Tipler-cylinder pair as a tunable time-travel harness. https://github.com/Zynerji/systrophe/blob/main/paper/systrophe_time_travel.pdf, 2026.
- [2] C. Knopp. Systrophē II: quantum field theory on a Tipler-pair background. https://github.com/Zynerji/systrophe/blob/main/paper/systrophe_qft_on_ctc.pdf, 2026.
- [3] C. Knopp. Clifford-structured Deutsch-CTC channels: empirical amplification, spectral oracle, and the encoding-dependence of chronology protection. https://github.com/Zynerji/systrophe/blob/main/paper/dctc_amplification.pdf, 2026.
- [4] C. Knopp. Deutsch-CTC is a limit-cycle theory: numerical evidence. https://github.com/Zynerji/systrophe/blob/main/paper/dctc_cycles.pdf, 2026.